Fiscal Year:	FY 2010	**Task Last Updated:**	FY 12/23/2010
PI Name:	McKinley, Gareth Ph.D.		
Project Title:	The Effects of Shear History on the Extensional Rheology of Complex Fluids		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	02139-4307	**Congressional District:**	8
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	03/08/2007	**End Date:**	03/07/2011
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
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Flight Program:	Shuttle/ISS		
Flight Assignment:	STS-120 ; STS-123 ; STS-126 ; STS-128 ISS NOTE: original end date was 3/7/2008; extended to 3/7/2009 in 03/2008; extended to 03/07/2011 in 06/2009 (10/2009)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Soulages, Johannes (Massachusetts Institute of Technology)		
Grant/Contract No.:	NNX07AE08G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The extensional viscosity is a fundamental physical property of all non-Newtonian materials, and cannot be determined from simple viscometric shear flow experiments. Constitutive equations for viscoelastic fluids such as dilute polymer solutions predict large changes in the extensional viscosity as the elongation rate is increased; however, the validity of these theories has only recently been confirmed due to paucity of experimental data that were available in extensional flows. To date, most quantitative rheological measurements in shear-free flows have only been possible for highly elastic or "stiff" materials such as polymer melts which can easily be elongated without sagging under a gravitational body force. In a review for the National Research Council on the future of Fluid Dynamics Research, Denn (1993) remarked that "...there are two major outstanding problems in rheological measurements. One is the measurement of extensional stresses of mobile fluids such as polymer solutions." Previous research on filament stretching rheometry supported by NASA in the P.I.'s lab has gone at least some way to addressing this situation. The perturbative effects of gravity and/or container boundaries that have interfered with almost all previous studies of extensional rheology in mobile fluids have been addressed in two ways; (i) by miniaturizing the apparatus or using neutral buoyancy tanks to minimize body forces, or (ii) by performing similar experiments in an extended microgravity environment. Using the instrumentation developed it has become possible to obtain accurate measurements of the extensional viscosity for more 'mobile' fluids such as dilute and concentrated polymer solutions, gels, suspensions and liquid crystalline materials. This rheological data is important in validating or extending constitutive models that aim to describe the response of complex fluids in arbitrary deformation fields. Such models will ultimately allow designers of both space- and ground-based material processes to use numerical simulations for a priori predictions of the properties of new materials that involve complex two- and three-dimensional fluid flows. The objectives of the proposed research are to (i) adapt an existing filament stretching device developed at MIT in order to enable extensional measurements of a wider range of complex fluids including worm-like micellar fluids and associative polymer gels; (ii) To combine these mechanical measurements with a microstructural probe of the evolution of the molecular orientation of the polymer chains using flow-induced birefringence; (iii) to explore, for the first time, the effects of a pre-shear history on the transient extensional viscosity in a uniaxial stretching flow for dilute polymer solutions together with the subsequent relaxation behavior after cessation of extensional deformation; (iv) to investigate the viscoelastic free-surface adhesion instabilities and the process of fibril formation in viscoelastic fluids which contribute significantly to the work of adhesion that must be input to separate two rigid planar substrates.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The SHERE Experiment is an ISS glovebox experiment designed to study the effect of preshear on the transient evolution of the viscoelastic tensile stresses for monodisperse dilute polymer solutions together with the subsequent fluid relaxation behavior after cessation of the extensional deformation. The combination of both rotational shearing and extensional stretching flows is common in many polymer processing operations. Therefore, a thorough understanding of the shear-free flows of polymeric liquids is required in order to accurately predict their extensional properties. This research has thus direct applications in many industrial processes such as film coating operations, stability and breakup of jets, enhanced oil recovery and turbulent drag reduction for advanced aircraft, boats, and submarines. In addition, understanding and quantifying the extensional rheology of complex fluids allows to improve and control earth-based manufacturing processes such as fiber micro/nanofabrication, atomization, electrospinning and to understand elastically-driven flow instabilities in microfluidics.
Task Progress:	Following the launch of the SHERE main hardware experiment on board Shuttle Mission STS-120 (10A) on October 22, 2007, a first batch of 20 fluid modules (FM) was launched on board Shuttle Mission STS-123 (1J/A) on March 11, 2008. Using the 20 FM for Increment 17, 28 experiments were performed, including 20 test points (TP) and 8 repeat test points using the previously deployed FM. A second batch of 25 redesigned fluid modules was subsequently launched on board Shuttle Mission STS-126 (ULF2) on November 14, 2008. They were used in 28 experiments during Increment 18, including 25 test points and 3 special test points whose purpose was to troubleshoot the electronics of the force transducer. The SHERE experiments were performed by astronaut Gregory Chamitoff during Increment 17 between July and September 2008 and by astronaut Mike Fincke during Increment 18 between December 2008 and January 2009. The data analysis is currently underway for the test points of the Increments 17 and 18. [Ed. note: compiled 12/23/10 by Task Book editor from unsubmitted report by PI in November 2009]
Bibliography Type:	Description: (Last Updated: 09/04/2025)
Abstracts for Journals and Proceedings	Hall NR, McKinley GH, Erni P, Soulages J, Magee KS. "Preliminary findings from the SHERE ISS experiment." Presented at the 47th AIAA Aerospace Sciences Meeting including The New Horizons Forum and Aerospace Exposition, Orlando, Florida, Jan. 5-8, 2009. 47th AIAA Aerospace Sciences Meeting including The New Horizons Forum and Aerospace Exposition, Orlando, Florida, Jan. 5-8, 2009. AIAA-2009-618 paper. , Jan-2009
Articles in Peer-reviewed Journals	Soulages J, Oliveira MSN, Sousa PC, Alves MA, McKinley GH. "Investigating the stability of viscoelastic stagnation flows in T-shaped microchannels." Journal of Non-Newtonian Fluid Mechanics, 2009 Nov;163(1-3): 9-24. http://dx.doi.org/10.1016/j.jnnfm.2009.06.002 , Nov-2009